

System Architecture: Skill-Based Hobby Recommendation System

Architecture Type:

Client-Side Single Page Application (SPA) with Hybrid Recommendation System.

Components:

1. UI Layer:

- Landing Page
- Quiz Engine (Adaptive)
- Results Page
- Detail Page

2. Logic Layer:

- User Vector Generator
- Recommendation Engine (Cosine Similarity)
- Hybrid Scoring (Similarity + Popularity)
- Confidence Scoring
- Personality Classification
- Explanation Engine

3. Data Layer:

- Static dataset (hobbies.js)
- Popularity scores
- Personality mapping rules

4. Optional AI Layer:

- Mood Detection (text input)
- ML Model (optional classification)

Flow:

User → Quiz → Vector → Hybrid Scoring → Ranking → Explanation → UI Output